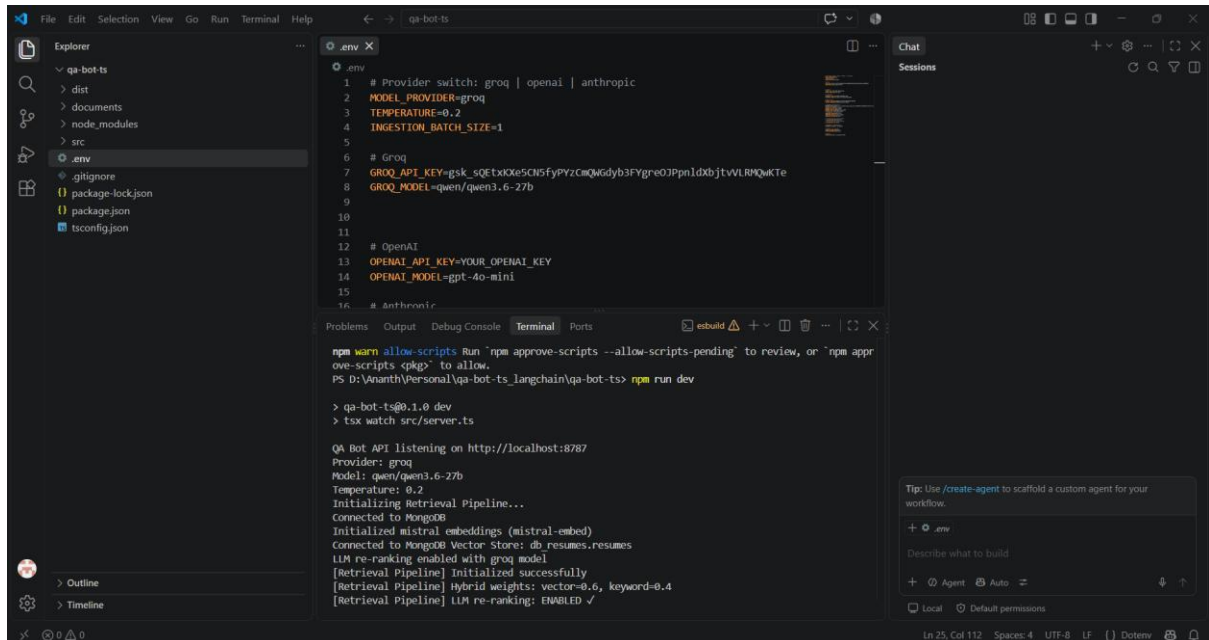


Lang Chain Memory:

Server Running:



The screenshot shows the Visual Studio Code interface for the 'qa-bot-ts' project. The Explorer on the left shows the project structure with files like .env, .gitignore, package-lock.json, package.json, and tsconfig.json. The .env file in the center contains configuration for Groq and OpenAI. The terminal at the bottom shows the command 'npm run dev' being executed, which starts the server on http://localhost:8787. The chat interface on the right is also visible.

```
1 # Provider switch: groq | openai | anthropic
2 MODEL_PROVIDER=groq
3 TEMPERATURE=0.2
4 INGESTION_BATCH_SIZE=1
5
6 # Groq
7 GROQ_API_KEY=gsk_5QEtX0KxSCH5FyPYZCmQWdyb3FYgre0JPpnlDxbJtVtVLRmQkTe
8 GROQ_MODEL=qwen/qwen3.6-72b
9
10
11
12 # OpenAI
13 OPENAI_API_KEY=YOUR_OPENAI_KEY
14 OPENAI_MODEL=gpt-4o-mini
15
16 # anthropic
```

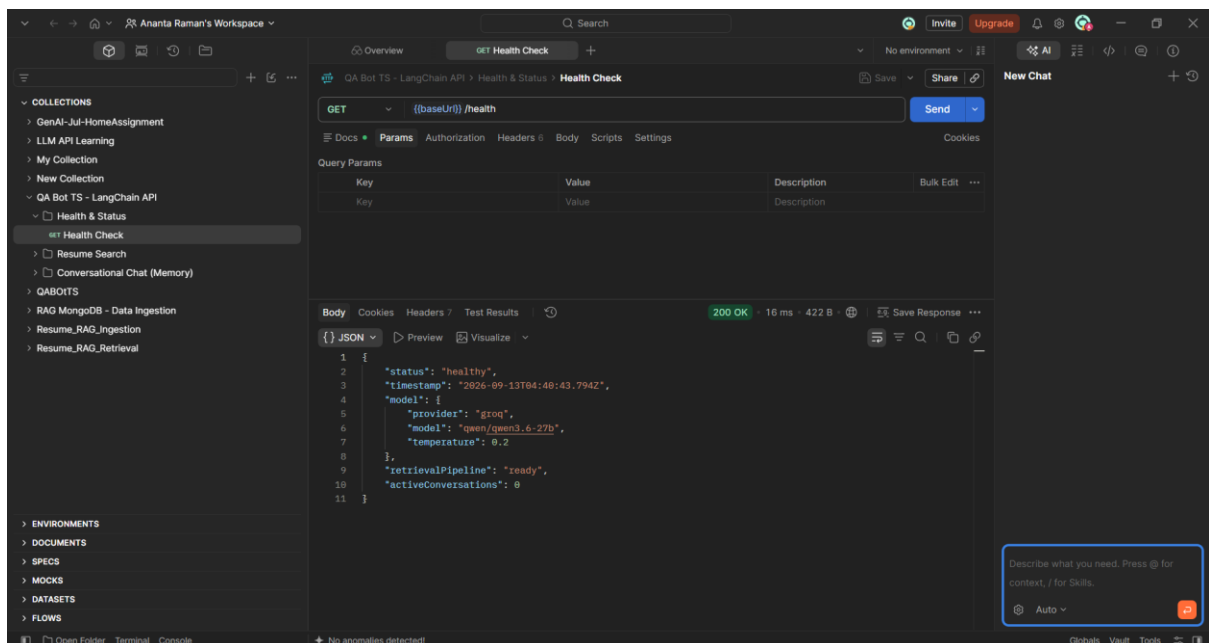
Terminal Output:

```
npm warn allow-scripts Run 'npm approve-scripts --allow-scripts-pending' to review, or 'npm approve-scripts --allow-scripts-pending' to allow.
PS D:\Ananth\Personal\qa-bot-ts_langchain\qa-bot-ts> npm run dev

> qa-bot-ts@0.1.0 dev
> tsx watch src/server.ts

QA Bot API listening on http://localhost:8787
Provider: groq
Model: qwen/qwen3.6-72b
Temperature: 0.2
Initializing Retrieval Pipeline...
Connected to MongoDB
Initialized mistral embeddings (mistral-embed)
Connected to MongoDB Vector Store: db.resumes.resumes
LUM re-ranking enabled with groq model
[Retrieval Pipeline] Initialized successfully
[Retrieval Pipeline] Hybrid weights: vector=0.6, keyword=0.4
[Retrieval Pipeline] LUM re-ranking: ENABLED ✓
```

Postman Health method:



The screenshot shows the Postman interface for the 'QA Bot TS - LangChain API' collection. The 'Health Check' endpoint is selected, and the response is shown in JSON format. The response indicates that the system is healthy and provides details about the provider, model, and retrieval pipeline.

```
1 {
2   "status": "healthy",
3   "timestamp": "2026-09-13T04:40:43.794Z",
4   "model": {
5     "provider": "groq",
6     "model": "qwen/qwen3.6-72b",
7     "temperature": 0.2
8   },
9   "retrievalPipeline": "ready",
10  "activeConversations": 0
11 }
```

Vector Search:

The screenshot shows the 'Vector Search (Semantic)' interface. The left sidebar lists various collections, including 'QA Bot TS - LangChain API' and 'Resume Search'. The main panel displays a POST request to '({baseUri})/search/resumes' with a JSON body:

```
{ 1: { 2: "query": "Selenium Java automation tester", 3: "searchType": "vector", 4: "topk": 5 5: }
```

. The response is a 200 OK status with a 1m 43.50s duration and 1.83 KB body. The JSON response includes a score of 0.8886578334434569 and a reasoning message: "Error during LLM analysis - score from vector search".

Body Cookies Headers Test Results 200 OK · 1 m 43.50 s · 1.83 KB Save Response

```
1 {
2   "query": "Selenium Java automation tester",
3   "searchType": "vector",
4   "topk": 5,
5   "resultCount": 5,
6   "duration": 183476,
7   "results": [
8     {
9       "name": "Theegela Sahitya Bharathi",
10      "fileName": "Sahitya Bharathi Theegela .pdf",
11      "email": "t.sahityabharathi24@gmail.com",
12      "phoneNumber": "+91-7993245519",
13      "content": "",
14      "fullContent": "",
15      "score": 0.8886578334434569,
16      "matchType": "vector",
17      "llmAnalysis": {
18        "reasoning": "Error during LLM analysis - score from vector search"
19      }
20    }
21  ]
22 }
```

Keyword Search:

The screenshot shows the 'Keyword Search (BM25-style)' interface. The left sidebar lists various collections, including 'QA Bot TS - LangChain API' and 'Resume Search'. The main panel displays a POST request to '({baseUri})/search/resumes' with a JSON body:

```
{ 1: { 2: "query": "Playwright test engineer", 3: "searchType": "keyword", 4: "topk": 5 5: }
```

. The response is a 200 OK status with a 5.17s duration and 1.81 KB body. The JSON response includes a score of 0.8886578334434569 and a reasoning message: "Error during LLM analysis - score from keyword search".

Body Cookies Headers Test Results 200 OK · 5.17 s · 1.81 KB Save Response

```
1 {
2   "query": "Playwright test engineer",
3   "searchType": "keyword",
4   "topk": 5,
5   "resultCount": 5,
6   "duration": 5143,
7   "results": [
8     {
9       "name": "PROFESSIONAL SUMMARY",
10      "fileName": "Arunkumar CV 1 (1).pdf",
11      "email": "arjun.arunkumar1968@gmail.com",
12      "phoneNumber": "9629583892",
13      "content": "",
14      "fullContent": "",
15      "score": 0.8886578334434569,
16      "matchType": "keyword",
17      "llmAnalysis": {
18        "reasoning": "Error during LLM analysis - score from keyword search"
19      }
20    }
21  ]
22 }
```

Hybrid Search:

The screenshot displays the 'Hybrid Search (Vector + Keyword)' configuration in the workspace. The search body is set to raw and contains the following JSON:

```
1 {
2   "query": "Automation engineer based out of chennai with 5 years of experience",
3   "searchType": "hybrid",
4   "topK": 10
5 }
```

The response is in JSON format and shows a 200 OK status. The response body is as follows:

```
1 {
2   "query": "Automation engineer based out of chennai with 5 years of experience",
3   "searchType": "hybrid",
4   "topK": 10,
5   "resultCount": 10,
6   "duration": 102655,
7   "results": [
8     {
9       "name": "PROFESSIONAL SUMMARY",
10      "fileName": "Naukri_AravindKumar[3y_4m].pdf",
11      "email": "aravindborn2001@gmail.com",
12      "phoneNumber": "+91 9566013221",
13      "content": "",
14      "fullContent": "",
15      "score": 0.8306269768572489,
16      "matchType": "hybrid",
17      "llmAnalysis": {
18        "reasoning": "Error during LLM analysis - score from vector search"
19      }
20    }
21  ]
22 }
```

New Conversation:

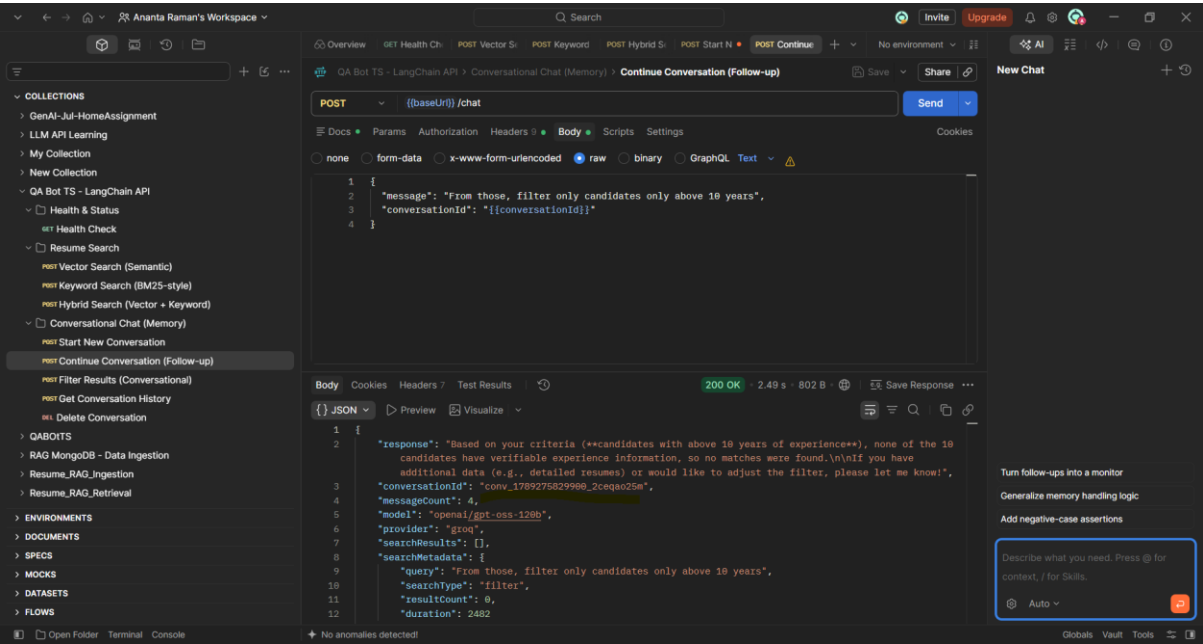
The screenshot displays the 'Start New Conversation' configuration in the workspace. The chat body is set to raw and contains the following JSON:

```
1 {
2   "message": "Find Selenium testers with more than 5 years of experience."
3 }
```

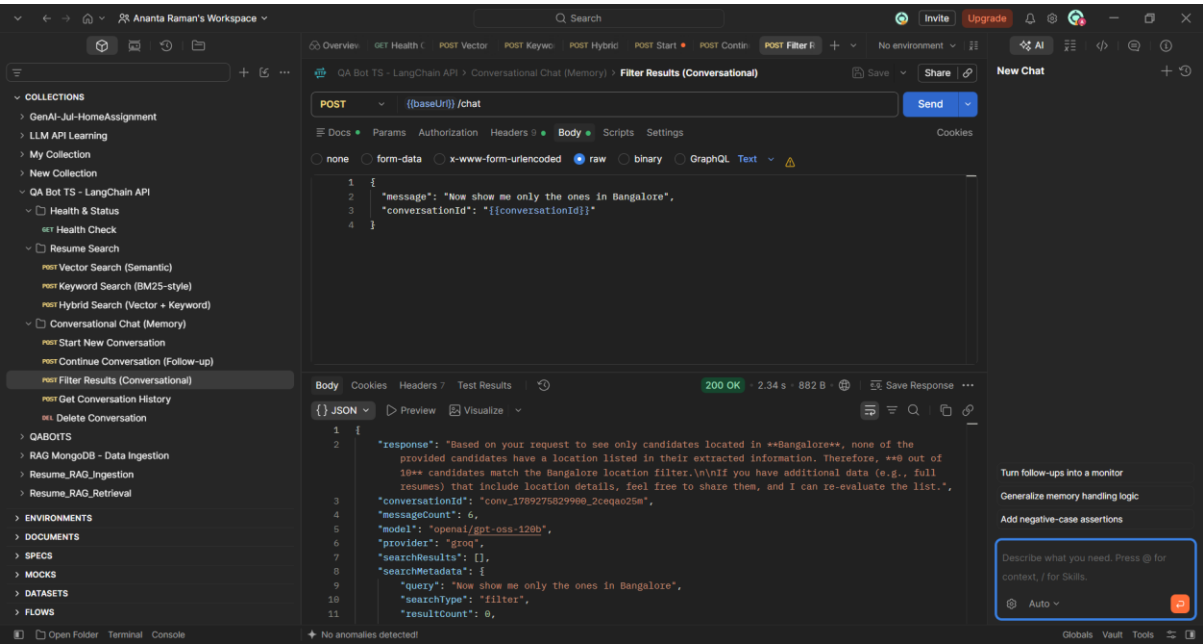
The response is in JSON format and shows a 200 OK status. The response body is as follows:

```
1 {
2   "response": "Answer\nBased on the information shown in the search results, none of the candidates explicitly list more than 5 years of Selenium testing experience in the preview snippets that were provided. The match scores (ranging from 0.83 to 0.60) indicate that these profiles are relevant to 'Selenium tester,' but the exact length of experience is not visible in the excerpts. What you can do next\n\n | Candidate (as shown) | Match Score | Contact (if you want to reach out) | why they're worth a deeper look |\n-----|\n | Highest relevance score - open the full resume to verify years of experience, |\n | **PROFESSIONAL SUMMARY** (Aravind) | 0.886 | aravindborn2001@gmail.com / +91 9566013221 | Strong match - likely has Selenium experience; check the detailed work history. |\n | **Profile Summary** (Amreen) | 0.707 | amreenbegam812@gmail.com / +91 9344374143 | Good relevance - review for experience length. |\n | **Santhana Bharathi** | 0.635 | santhanabharathi2001@gmail.com | Moderate match - could meet the 5-year threshold. |\n | **MK** | 0.626 | mukeshkanna2395.m@gmail.com / 8056375649 | Moderate match - worth opening. |\n | **KARTHIK S** | 0.625 | karthikasahithi2506@gmail.com | Moderate match - review details. |\n | **Science In Vids** | 0.609 | +91 8778351504 | Lower match - still a candidate to examine. |\n | **Srividhya V** | 0.607 | Srividhya.maniam@gmail.com / +91 8070003334 | Lower match - check for experience length. |\n | **MOHAN K** | 0.605 | cmohankm@gmail.com / 9677756330 | Lower match - open to verify. |\n | **Theegela Sahithya Bharathi** | 0.605 | t.sahithyabharathi24@gmail.com / +91 7993245519 | Lower match - open to verify. |\n\nNext steps\n\n | Open the full resumes for the top-scoring
```

Conversation continuation:



Filter Results:



Conversation History:

The screenshot displays the Ananta Raman's Workspace interface. The left sidebar shows a collection of projects, including 'QA Bot TS - LangChain API'. The main panel is titled 'Get Conversation History' and shows a POST request to the endpoint '({baseUrl}) /chat/history'. The request body is a JSON object with a 'conversationId' field. The response is a 200 OK status with a 9 ms response time and a 3.74 KB body. The response body is a JSON object with a 'conversationId' field and a 'messages' array containing two messages: a user message and an assistant message. The assistant message content is: '***Answer***\nBased on the information shown in the search results, none of the candidates explicitly list *more than 5 years of Selenium testing experience* in the preview snippets that were provided. The match scores (ranging from 0.83 to 0.66) indicate that these profiles are relevant to "Selenium tester," but the exact length of experience is not visible in the excerpts.\n\nWhat you can do next*\n\n# | Candidate (as shown) |

Deletion of Conversation:

The screenshot displays the Ananta Raman's Workspace interface. The left sidebar shows a collection of projects, including 'QA Bot TS - LangChain API'. The main panel is titled 'Delete Conversation' and shows a DELETE request to the endpoint '({baseUrl}) /chat/({conversationId})'. The response is a 200 OK status with a 5 ms response time and a 348 B body. The response body is a JSON object with a 'success' field set to true, a 'conversationId' field, and a 'message' field: 'Conversation deleted successfully'.

Again, to retrieve Conversation History:

The screenshot displays the Ananta Raman's Workspace interface. On the left, a sidebar lists various collections under 'COLLECTIONS', including 'GenAI-Jul-HomeAssignment', 'LLM API Learning', 'My Collection', 'New Collection', 'QA Bot TS - LangChain API', 'Health & Status', 'Resume Search', 'Vector Search (Semantic)', 'Keyword Search (BM25-style)', 'Hybrid Search (Vector + Keyword)', 'Conversational Chat (Memory)', 'Start New Conversation', 'Continue Conversation (Follow-up)', 'Filter Results (Conversational)', 'Get Conversation History', and 'Delete Conversation'. The 'Get Conversation History' endpoint is selected.

The main area shows a REST client request to `{{baseUri}}/chat/history` with a `POST` method. The request body is a JSON object:

```
{ 1: { 2: "conversationId": "{{conversationId}}" 3: }}
```

. The response is a `404 Not Found` error with a status of `404`, a message of `"error": "Conversation not found: conv_1789275829968_2ceqao25m"`, and a status bar indicating `Investigate 404 error`.

On the right, a 'New Chat' panel is visible, showing a message: 'Investigate 404 error. The request is failing with a 404 error because the conversation ID provided in the request body does not exist. Please check the conversation ID or verify if a new conversation needs to be created first.' Below this, there are buttons for 'Open Source Code' and 'Think Harder'.